

ROOT CAUSE ANALYSIS

DVWA Domain Access Failure Through SafeLine WAF

Traefik TCP/443 Listener Conflict

Status	RESOLVED
Incident date	22 September 2026
Environment	Kali Linux client + Ubuntu homelab host + k3s/Traefik + SafeLine + DVWA
Primary root cause	TCP/443 was already handled by Traefik, so SafeLine did not receive the request.
Final working path	https://dvwa.local:8443/DVWA/login.php

Prepared as a technical incident record and homelab troubleshooting reference.

1. Executive Summary

DVWA was reachable directly at <http://192.168.1.17:8080/DVWA/login.php>, but the intended protected domain <https://dvwa.local/DVWA/login.php> returned an HTTP 404 response.

Troubleshooting established that the DVWA backend was healthy and that Kali could resolve dvwa.local to 192.168.1.17. The decisive evidence came from a verbose TLS request: the server on port 443 presented a certificate with CN=TRAEFIK DEFAULT CERT. This proved the HTTPS request was terminating at Traefik, not SafeLine. Because Traefik had no matching route for dvwa.local, it returned its default 404 response.

SafeLine was then moved to HTTPS port 8443 while retaining its upstream as <http://192.168.1.17:8080>. Re-testing the same hostname and application path on port 8443 returned HTTP 200, presented the dvwa.local certificate, and issued an sl-session cookie, confirming that the request was now being processed through SafeLine and successfully proxied to DVWA.

Root cause in one sentence

SafeLine and Traefik were intended to use the same host-side HTTPS port, but TCP/443 was already being handled by Traefik; therefore requests to <https://dvwa.local> reached Traefik instead of SafeLine and resulted in a 404.

2. Impact and Scope

- Impact was limited to the homelab access path for DVWA through the intended SafeLine domain.
- DVWA itself remained available through direct backend access on 192.168.1.17:8080.
- No evidence indicated an Apache or DVWA application failure.
- Existing Traefik/k3s services were left intact; Traefik was not disabled as part of the fix.
- The incident exposed an architectural port ownership conflict on a single Ubuntu host running both the k3s/Traefik homelab and SafeLine.

3. Intended and Actual Traffic Flow

3.1 Intended flow before troubleshooting

```
Kali Client
| HTTPS :443
v
dvwa.local / 192.168.1.17
|
v
SafeLine WAF
| HTTP :8080
v
Apache / DVWA
```

3.2 Actual failing flow

```

Kali Client
| HTTPS :443
v
192.168.1.17:443
|
v
Traefik
| no matching route for Host: dvwa.local
v
HTTP 404

```

3.3 Working flow after remediation

```

Kali Client
| HTTPS :8443
v
SafeLine WAF
| reverse proxy
v
http://192.168.1.17:8080
|
v
Apache / DVWA
|
v
HTTP 200

```

4. Troubleshooting Timeline and Isolation Logic

Step	Test / Observation	Finding	Conclusion
1	Direct backend test: curl -I http://192.168.1.17:8080/DVWA/login.php	HTTP/1.1 200 OK; Apache responded	DVWA, Apache, port 8080, and basic network path were healthy.
2	Checked Kali hostname resolution for dvwa.local	Kali initially lacked the required host mapping	Added 192.168.1.17 dvwa.local to Kali /etc/hosts.
3	Retested domain over HTTPS	Hostname resolved, but request still returned 404	DNS/hosts was no longer the primary failure.
4	Ran curl -k -v against https://dvwa.local	TLS certificate showed CN=TRAEFIK DEFAULT CERT	Port 443 traffic was reaching Traefik, not SafeLine.
5	Reviewed port ownership and service design	Traefik already used the standard HTTPS listener on the same host	SafeLine could not be the effective endpoint for dvwa.local:443 in the current layout.
6	Changed SafeLine application listener to 8443	SafeLine kept upstream http://192.168.1.17:8080	Separated the two ingress listeners without disabling Traefik.

Step	Test / Observation	Finding	Conclusion
7	Ran curl -k -v against https://dvwa.local:8443	CN=dvwa.local, HTTP/1.1 200 OK, sl-session cookie	SafeLine path confirmed working end-to-end.

5. Technical Evidence

5.1 Backend health check - successful

```
$ curl -I http://192.168.1.17:8080/DVWA/login.php
HTTP/1.1 200 OK
Server: Apache/2.4.58 (Ubuntu)
Content-Type: text/html; charset=utf-8
```

Interpretation: the application backend was serving DVWA correctly. This removed Apache, DVWA, and port 8080 availability from the primary fault domain.

5.2 Failed domain request on default HTTPS port 443

```
$ curl -k -v https://dvwa.local/DVWA/login.php
* Host dvwa.local:443 was resolved.
* IPv4: 192.168.1.17
* Trying 192.168.1.17:443...
* Server certificate:
* subject: CN=TRAEFIK DEFAULT CERT <<< KEY EVIDENCE
* issuer: CN=TRAEFIK DEFAULT CERT
* Established connection to dvwa.local (192.168.1.17 port 443)
> GET /DVWA/login.php HTTP/2
> Host: dvwa.local
< HTTP/2 404

404 page not found
```

Why this log mattered

The hostname resolved correctly, but the certificate identified Traefik as the TLS endpoint on TCP/443. That is the key observation that shifted the investigation from DNS/DVWA to listener ownership and reverse-proxy routing.

5.3 Successful SafeLine request on port 8443

```
$ curl -k -v https://dvwa.local:8443/DVWA/login.php
* Host dvwa.local:8443 was resolved.
* IPv4: 192.168.1.17
* Trying 192.168.1.17:8443...
* Server certificate:
* subject: ... CN=dvwa.local <<< EXPECTED CERTIFICATE
* Established connection to dvwa.local (192.168.1.17 port 8443)
> GET /DVWA/login.php HTTP/1.1
> Host: dvwa.local:8443
< HTTP/1.1 200 OK
< Set-Cookie: sl-session=...; Secure; Path=/; HttpOnly

<title>Login :: Damn Vulnerable Web Application (DVWA)</title>
```

Interpretation: changing only the destination listener from 443 to 8443 caused the request to reach the intended SafeLine endpoint. The HTTP 200 response confirmed successful reverse proxying to DVWA, while the SafeLine session cookie provided additional evidence that the WAF processed the request.

6. Root Cause Analysis

6.1 Primary root cause

Port/listener ownership conflict on the shared Ubuntu host. Traefik was already the effective HTTPS listener on TCP/443. SafeLine was initially configured for the same public-facing port on the same host. Requests to `https://dvwa.local` therefore terminated at Traefik, not SafeLine. Because Traefik had no matching route for `dvwa.local`, it returned its default HTTP 404 response.

6.2 Contributing factor

Kali initially did not have the required `dvwa.local` host mapping. Adding `192.168.1.17 dvwa.local` corrected name resolution, but the 404 persisted. This was a contributing configuration issue, not the final root cause.

6.3 Why the 404 was misleading

An HTTP 404 can be generated by an upstream application, a reverse proxy, an ingress controller, or another intermediary. The status code alone did not identify the responding component. Inspecting the TLS certificate with `curl -v` revealed that the service answering on port 443 was Traefik. This was the decisive discriminator.

6.4 Five-Whys summary

Question	Answer
Why did <code>https://dvwa.local</code> fail?	The request returned HTTP 404.
Why was a 404 returned?	The request reached Traefik, which had no route for <code>dvwa.local</code> .
Why did it reach Traefik instead of SafeLine?	The browser used the default HTTPS port 443, and Traefik was already handling that listener.
Why could SafeLine not transparently own the same endpoint?	Both services were colocated on the same Ubuntu host and competed for the same IP/port combination.
Why was this not obvious initially?	The hostname resolved successfully and the response was a valid HTTP 404; only TLS endpoint inspection exposed which reverse proxy answered.

7. Corrective Action and Current Stable State

1. Keep Traefik enabled on TCP/443 for the existing k3s homelab ingress.
2. Change the SafeLine protected application listener from HTTPS/443 to HTTPS/8443.
3. Retain the SafeLine upstream as `http://192.168.1.17:8080` so SafeLine reverse-proxies to DVWA over the backend port.
4. Keep Kali host resolution pointing `dvwa.local` to `192.168.1.17` while both services remain on the same host.
5. Use `https://dvwa.local:8443/DVWA/login.php` when explicitly testing the SafeLine-protected path.

Current port allocation

Port	Component	Purpose
443	Traefik	Standard HTTPS ingress for k3s/homelab services

8080	Apache / DVWA	DVWA backend
8443	SafeLine WAF	Protected DVWA HTTPS frontend
9443	SafeLine	Management interface

8. Recommendations and Future Plan

8.1 Immediate / keep as the known-good baseline

- Keep Traefik on 443 and SafeLine on 8443 while the WAF lab is still being built and validated.
- Document both access paths clearly: direct backend for controlled troubleshooting and SafeLine path for protected testing.
- Do not treat 8443 as inherently less secure than 443. Security depends on TLS configuration, certificates, firewalling, and application/WAF controls; 443 is primarily the standard HTTPS convention.
- Retain the A/B curl commands as a repeatable health check after future configuration changes.

8.2 Near-term improvement - clean URL using Traefik as a TCP router

A future improvement can preserve the clean URL `https://dvwa.local` while keeping the single Ubuntu host. Traefik would continue owning TCP/443 but would route connections for the `dvwa.local` TLS SNI to SafeLine on 8443. The preferred lab design is TLS passthrough so SafeLine, not Traefik, terminates the DVWA TLS session and performs WAF inspection.

```
Client
| https://dvwa.local (TCP 443)
v
Traefik :443
| SNI match: dvwa.local
| TLS passthrough
v
SafeLine :8443
| TLS termination + WAF inspection
v
DVWA :8080
```

Important design note

Traefik in this model acts as the connection traffic director. SafeLine remains the security enforcement point for DVWA. Validate that the configured Traefik TCP route uses passthrough as intended before describing the architecture as end-to-end WAF inspection.

8.3 Long-term architecture - dedicated SafeLine IP or VM

The cleanest production-like topology is to give SafeLine a separate network identity, such as another VM or dedicated IP, so both Traefik and SafeLine can use standard TCP/443 without competing for the same IP/port tuple.

```
192.168.1.17:443 -> Traefik / k3s
192.168.1.18:443 -> SafeLine WAF
SafeLine upstream -> 192.168.1.17:8080 (DVWA)
```

A second physical server is not strictly required to understand the concept: one host can have multiple IP addresses. However, with the current k3s/Traefik networking, simply adding a secondary IP may not be sufficient if Traefik is

effectively bound to all local addresses on port 443. A separate VM/network namespace or carefully designed service-specific binding would be cleaner.

8.4 Security hardening after the routing design is stable

- Restrict direct client access to DVWA port 8080 so normal application traffic cannot bypass SafeLine. Keep a controlled administrative/test exception only if needed.
- Use host-based firewall rules to allow only the intended SafeLine-to-DVWA backend path once the lab architecture is finalized.
- Replace or trust the lab certificate appropriately if browser certificate warnings are no longer desired. Do not use -k as a normal production validation practice; it is acceptable here only for the self-signed lab certificate.
- Keep SafeLine management (9443) restricted to the management network or trusted admin hosts where practical.
- Enable and retain SafeLine access/security logs, plus Traefik logs for the edge routing layer, so future path failures can be correlated quickly.
- Document which component owns each host port and which hostnames are routed by Traefik versus SafeLine.

9. Post-Change Validation Checklist

Check	Command / Action	Expected Result
Hostname resolution	getent hosts dvwa.local	Returns the expected homelab IP.
Backend health	curl -I http://192.168.1.17:8080/DVWA/login.php	HTTP 200/302 from Apache/DVWA.
Traefik default path test	curl -k -v https://dvwa.local/DVWA/login.php	Until a Traefik route is added, shows Traefik certificate / 404 as expected.
SafeLine protected path	curl -k -v https://dvwa.local:8443/DVWA/login.php	HTTP 200 and dvwa.local certificate; SafeLine session evidence present.
Browser test	Open https://dvwa.local:8443/DVWA/login.php	DVWA login page loads through SafeLine.
Future TLS passthrough test	curl -k -v https://dvwa.local/DVWA/login.php	After implementation, connects on 443 but presents the SafeLine/dvwa.local certificate and returns HTTP 200.

10. Key Lessons Learned

- A successful DNS lookup only proves name resolution; it does not prove the request reaches the intended reverse proxy or application.
- A 404 response does not identify the failing layer. Always determine which component generated the response.
- Verbose TLS inspection can identify the actual endpoint. In this incident, CN=TRAEFIK DEFAULT CERT was the critical clue.
- On a shared host, the combination of IP address + protocol + port must have clear ownership. Multiple ingress products cannot independently own the same effective listener without an explicit routing design.
- Changing one variable at a time is powerful: hostname, IP, and application path stayed the same, while changing 443 to 8443 changed the responding endpoint from Traefik/404 to SafeLine/200.
- Non-standard HTTPS ports such as 8443 are valid for labs and internal services. Port 443 is preferred for user-facing convenience and standardization, not because the port number itself provides stronger encryption.

11. Final RCA Statement

Final statement

DVWA domain access failed because HTTPS requests to dvwa.local implicitly used TCP/443, which was already being handled by Traefik on the shared Ubuntu host. The verbose curl log exposed this through the certificate subject CN=TRAEFIK DEFAULT CERT. Traefik had no route for dvwa.local and returned HTTP 404. Name resolution on Kali was also corrected, but it was not the final cause. The incident was resolved by moving the SafeLine application listener to TCP/8443 while retaining its upstream at http://192.168.1.17:8080. Validation on 8443 returned HTTP 200, the expected dvwa.local certificate, and a SafeLine session cookie. The recommended next evolution is to keep this stable baseline, then optionally configure Traefik TCP/SNI routing with TLS passthrough from 443 to SafeLine 8443 for a clean URL, and ultimately move SafeLine to a dedicated IP/VM if a more production-like edge architecture is desired.

Appendix A - Repeatable Diagnostic Commands

```
# Verify resolution
getent hosts dvwa.local

# Verify direct DVWA backend
curl -I http://192.168.1.17:8080/DVWA/login.php

# Inspect the service answering default HTTPS/443
curl -k -v https://dvwa.local/DVWA/login.php

# Inspect the SafeLine-protected path
curl -k -v https://dvwa.local:8443/DVWA/login.php

# Review local listeners on Ubuntu
sudo ss -lntp | grep -E ":443|:8080|:8443|:9443"

# Review Traefik/k3s exposure
kubectl -n kube-system get svc traefik -o wide
kubectl -n kube-system get pods -o wide | grep -E "traefik|svclb"
```